YouTube Transcript API

Setup and Usage Guide

Run the Transcript-YouTube server on your own computer

Edition	Platforms covered
Part 1	macOS (Terminal)
Part 2	Windows 10 / 11 (PowerShell or Command Prompt)
Part 3	Using the API (Postman, curl, browser) and troubleshooting

Who is this for? Anyone who has received the Transcript-YouTube project folder and wants to run it locally to fetch YouTube video transcripts. No prior programming experience is required; just follow the steps in order.

Overview

Transcript-YouTube is a small web service built with Python (FastAPI, served by Uvicorn). Once started, it listens on your own computer at **http://127.0.0.1:8000** (also reachable as **http://localhost:8000**). You send it a YouTube link and it returns the video's transcript.

Setting it up takes five stages on either operating system:

#	Stage	What it does
1	Check Python	Confirms Python 3.9 or newer is installed.
2	Open the project folder	Moves your terminal into the Transcript-YouTube directory.
3	Create a virtual environment	Creates an isolated space (venv) so the project's libraries do not affect the rest of your computer.
4	Install libraries	Installs everything listed in requirements.txt.
5	Start the server	Runs the API on port 8000.

Before you begin

- You have the **Transcript-YouTube** project folder (it must contain **main.py** and **requirements.txt**).
- You have an internet connection (needed to install libraries and to fetch transcripts).
- You can open a terminal: **Terminal** on macOS, **PowerShell** or **Command Prompt** on Windows.

> **Tip:** Lines that start with **//** or are shown on a grey background are commands. Type or paste only the command itself, one line at a time, and press Enter after each.

Part 1: macOS

Step 1. Check Python

Open **Terminal** (press Cmd + Space, type "Terminal", press Enter) and run:

```
python3 --version
```

- If it prints **Python 3.x.x** and the version is **3.9 or higher**, continue to Step 2.
- If you see an error, or a dialog asks you to install the **Command Line Tools**, click **Install** and wait for it to finish, then run the command again.

Alternatively, install Python with Homebrew:

```
brew install python
```

If you do not have Homebrew, download the official installer from **https://www.python.org/downloads/macos/**, run it, then reopen Terminal and repeat the version check.

Step 2. Go to the project folder

Use **cd** (change directory) to enter the folder, then **ls** to list its contents:

```
cd ~/path/to/Transcript-YouTube
ls
```

Replace **~/path/to/Transcript-YouTube** with the real location of the folder. You should see **main.py** and **requirements.txt** in the list.

> **Tip:** Shortcut: type **cd** (with a trailing space) in Terminal, then drag the project folder from Finder into the Terminal window and press Enter.

Step 3. Create a virtual environment

Create the environment and activate it:

```
python3 -m venv venv
source venv/bin/activate
```

After activation your prompt starts with **(venv)**. This means the environment is active. You only need to create it once, but you must activate it every time you open a new Terminal window.

Step 4. Install the libraries

With the environment active, run:

```
pip install -r requirements.txt
```

Wait until the installation finishes without errors. This can take a minute or two.

Step 5. Start the server

Run:

```
python -m uvicorn main:app --host 127.0.0.1 --port 8000
```

- When you see **Uvicorn running on http://127.0.0.1:8000**, the server is live.
- **Keep this Terminal window open.** Closing it shuts the server down.
- To test it quickly, open **http://localhost:8000/docs** in your browser. This is the built-in Swagger interface, where you can try the API with no extra tools.

Running it again later (macOS)

A. The server is not running (fresh start):

```
cd ~/path/to/Transcript-YouTube
source venv/bin/activate
python -m uvicorn main:app --port 8000
```

B. The server is already running and you want to restart it:

- Click the Terminal window that is running the server.
- Press **Control + C** to stop it.

```
source venv/bin/activate
python -m uvicorn main:app --port 8000
```

C. You edited requirements.txt: reinstall the libraries before starting again.

```
source venv/bin/activate
pip install -r requirements.txt
python -m uvicorn main:app --port 8000
```

> **Note:** If you are still in the same Terminal window after pressing Control + C, you are already inside the project folder and the environment is still active, so the **cd** and **source** lines can be skipped. In a brand-new Terminal window, always run them first.

Part 2: Windows 10 / 11

You can use either **PowerShell** or **Command Prompt (cmd)**. The steps are the same except for activating the virtual environment (Step 3). Open one by pressing the Windows key, typing "PowerShell" or "cmd", and pressing Enter.

Step 1. Check Python

Run:

```
python --version
```

- If it prints **Python 3.x.x** with version **3.9 or higher**, continue to Step 2. (You can also try **py --version**.)
- If you get an error, or Windows opens the Microsoft Store, Python is not installed correctly.

To install Python:

- Download the installer from **https://www.python.org/downloads/windows/**.
- Run it. On the first screen, **tick "Add python.exe to PATH"** before clicking **Install Now**.
- When it finishes, **close and reopen** PowerShell or cmd, then run **python --version** again.

> **Note:** Forgetting to tick "Add python.exe to PATH" is the most common cause of "python is not recognized" errors. If that happens, re-run the installer, choose **Modify** or **Repair**, and enable the option, or uninstall and reinstall.

Step 2. Go to the project folder

Use **cd** to enter the folder and **dir** to list its contents:

```
cd C:\path\to\Transcript-YouTube
dir
```

Replace the path with the real location of the folder. You should see **main.py** and **requirements.txt**. If the folder is on a different drive (for example D:), switch drive first in cmd by typing **D:**, or use **cd /d D:\path\to\Transcript-YouTube**.

> **Tip:** In File Explorer, open the project folder, click the address bar, type **powershell** (or **cmd**), and press Enter. A terminal opens already inside that folder.

Step 3. Create a virtual environment

Create the environment (same in both shells):

```
python -m venv venv
```

Then activate it. Use the command that matches your shell:

Shell	Activation command
PowerShell	venv\Scripts\Activate.ps1
Command Prompt (cmd)	venv\Scripts\activate.bat

After activation your prompt starts with **(venv)**.

> **Note: PowerShell error "running scripts is disabled on this system"?** Run the following once, answer **Y** if asked, then try activating again:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Or simply use Command Prompt (cmd) with **activate.bat**, which has no such restriction.

Step 4. Install the libraries

With the environment active, run:

```
pip install -r requirements.txt
```

Wait until it finishes without errors.

Step 5. Start the server

Run:

```
python -m uvicorn main:app --host 127.0.0.1 --port 8000
```

- When you see **Uvicorn running on http://127.0.0.1:8000**, the server is live.
- **Keep this window open.** Closing it stops the server.
- Open **http://localhost:8000/docs** in your browser to test with the built-in Swagger interface.
- If Windows Defender Firewall asks whether to allow Python, you can click **Cancel** or allow on **Private networks**. Because the server only listens on 127.0.0.1, it is reachable only from your own computer.

Running it again later (Windows)

A. The server is not running (fresh start):

```
cd C:\path\to\Transcript-YouTube
venv\Scripts\Activate.ps1        (PowerShell)
venv\Scripts\activate.bat         (cmd)
python -m uvicorn main:app --port 8000
```

B. The server is already running and you want to restart it: click the terminal window running the server and press **Ctrl + C** to stop it (press it twice if it does not stop). Then:

```
venv\Scripts\Activate.ps1        (PowerShell)
venv\Scripts\activate.bat         (cmd)
python -m uvicorn main:app --port 8000
```

C. You edited requirements.txt: reinstall the libraries before starting again.

```
venv\Scripts\Activate.ps1        (PowerShell)
venv\Scripts\activate.bat         (cmd)
pip install -r requirements.txt
python -m uvicorn main:app --port 8000
```

> **Note:** Run only the activation line for your own shell, not both. The text in parentheses is a label, so do not type it.

Part 3: Using the API

These steps are the same on macOS and Windows, except where noted. The server must be running first.

Option 1: Swagger interface (easiest)

Open **http://localhost:8000/docs** in any browser. Expand the transcript endpoint, click **Try it out**, edit the JSON, and click **Execute**.

Option 2: Postman

- Open Postman and click **Import** (or press Cmd/Ctrl + O).
- Choose the **Raw text** tab and paste the curl command below.
- Click **Continue**, then **Import**, then **Send**.

```
curl -X POST http://localhost:8000/api/transcript \
  -H "Content-Type: application/json" \
  -d '{"url": "https://www.youtube.com/watch?v=dQw4w9WgXcQ",
       "timestamp": true, "mergeSentences": true}'
```

Or build the request by hand: set the method to **POST**, the URL to **http://localhost:8000/api/transcript**, then open **Body**, select **raw** and **JSON**, and paste:

```
{
  "url": "https://www.youtube.com/watch?v=dQw4w9WgXcQ",
  "timestamp": true,
  "mergeSentences": true
}
```

Request fields

Field	Type	Meaning
url	text	The full YouTube video link.
timestamp	true / false	Include timestamps with the transcript.
mergeSentences	true / false	Join transcript fragments into complete sentences.

Option 3: curl in a terminal

macOS (Terminal) - use the same command as above (single quotes around the JSON work in zsh/bash).

Windows (Command Prompt) - cmd does not accept single quotes, so wrap the JSON in double quotes and escape the inner ones:

```
curl -X POST http://localhost:8000/api/transcript ^
  -H "Content-Type: application/json" ^
  -d "{\"url\": \"https://www.youtube.com/watch?v=dQw4w9WgXcQ\", ^
  \"timestamp\": true, \"mergeSentences\": true}"
```

The **^** at the end of a line tells cmd the command continues on the next line. In cmd, a line ending in **^** may need to be the very last character (no trailing spaces).

> **Note:** In **PowerShell**, **curl** is an alias for a different command (Invoke-WebRequest) and will fail with these options. Type **curl.exe** instead, and prefer Swagger or Postman for JSON bodies, since PowerShell quoting is tricky.

Option 4: Chrome (or any browser)

Open this address:

```
http://localhost:8000/
```

This opens the server's root page in the browser. Use **/docs** for the interactive API tester.

Troubleshooting

Problem	Likely cause and fix
python3 / python: command not found (or "not recognized")	Python is missing or not on PATH. Install it (see Step 1 for your OS). On Windows, reinstall with "Add python.exe to PATH" ticked and reopen the terminal.
No (venv) at the start of the prompt	The environment is not active. Run the activation command for your OS and shell again.
No module named uvicorn / fastapi	Libraries are not installed in the active environment. Activate venv, then run **pip install -r requirements.txt**.
Could not open requirements file / Error loading ASGI app	You are in the wrong folder. Use **cd** to enter the project folder that contains main.py and requirements.txt.
Address already in use (port 8000)	Another program (or an old server) uses port 8000. Stop the old server with Control/Ctrl + C, or start on another port, e.g. **--port 8001**, and use that port in all URLs.
Windows: running scripts is disabled	In PowerShell run **Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned**, or use Command Prompt with activate.bat.
Browser cannot connect to localhost:8000	The server is not running or its terminal was closed. Start it again and wait for the "Uvicorn running" message.

Problem	Likely cause and fix
Transcript request returns an error	The video may have no captions, be private or age-restricted, or the link may be wrong. Try another public video that has captions.

Quick command reference

Task	macOS	Windows
Check Python	python3 --version	python --version
Create environment	python3 -m venv venv	python -m venv venv
Activate environment	source venv/bin/activate	PowerShell: venv\Scripts\Activate.ps1 cmd: venv\Scripts\activate.bat
Install libraries	pip install -r requirements.txt	pip install -r requirements.txt
Start server	python -m uvicorn main:app --port 8000	python -m uvicorn main:app --port 8000
Stop server	Control + C	Ctrl + C
Leave environment	deactivate	deactivate